Toward an Emergent Gravitational and Propagation Sector in General Emergence Mechanics

Persistence Depth, Shell Causal Speed, Refractive Gravity, and Weak-Field Schwarzschild Structure

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Abstract

This paper develops a unified weak-field programme for gravity and propagation in General Emergence Mechanics (GEM). The central proposal is that persistent physical structure exists within a dynamical coherence shell whose local depth modulates both restoring behaviour and signal propagation. Starting from an effective shell Lagrangian, we identify a natural shell propagation speed $c^2_{\text{shell}} = \lambda/\chi$, where λ is the spatial gradient stiffness and χ is the temporal inertia coefficient. We then introduce a class of depth-weighted effective potentials for which the local curvature at the preferred minimum gives a depth-dependent stiffness $\kappa_{\text{eff}}(x) = D(x)$ in a natural normalisation. Promoting the response coefficients to depth-dependent functions yields a variable propagation structure, from which Newtonian-like attraction follows through a refractive or travel-time argument. A further weak-field extension with separate temporal and spatial response functions yields a Schwarzschild-like metric form, providing a route to gravitational redshift and the full weak-field light-bending structure. The paper does not claim a completed derivation of Einstein's field equations or of the exact numerical value of the speed of light. Rather, it organises a mathematically explicit and internally connected programme in which gravity and propagation emerge from the same persistence structure.

Keywords: General Emergence Mechanics, GEM, emergent gravity, coherence shell, propagation speed, refractive gravity, persistence depth, weak-field metric, gravitational redshift, light bending

Note. This paper presents an exploratory weak-field framework rather than a finished gravitational theory. Its goal is to formalise the strongest currently available routes from shell dynamics to propagation and gravity.

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1 Introduction

General Emergence Mechanics (GEM) approaches stable existence through persistence rather than through geometry alone. In GEM, long-lived structures remain viable only when they stay coherent inside a dynamical shell governed by depth, stiffness, and relaxation (viscosity).

The present paper focuses on a specific possibility: that the same shell structure responsible for persistence may also determine both the characteristic speed of coherent propagation and the weak-field behaviour ordinarily attributed to gravity. In this picture, gravity is not assumed first as a force or curvature law. Instead, one asks whether local shell depth can simultaneously modify restoring behaviour and propagation structure, giving rise to Newtonian-like attraction and weak-field Schwarzschild features as emergent consequences.

2 The Effective Shell Lagrangian

Near the preferred phase minimum, the effective shell Lagrangian takes the form

$$\mathcal{L} = \frac{\chi}{2}(\partial_t \phi)^2 - \frac{\lambda}{2}(\nabla \phi)^2 - V(\phi),$$

where ϕ is the phase field with preferred minimum at $\phi_0 = 255°$, χ is the temporal inertia coefficient, λ is the spatial gradient stiffness, and $V(\phi)$ stabilises the preferred shell phase.

Linearising around the preferred minimum gives the wave equation

$$\chi \partial_t^2 \delta\phi - \lambda \nabla^2 \delta\phi + \kappa_{\text{eff}} \delta\phi = 0,$$

with dispersion relation

$$\omega^2 = \frac{\lambda}{\chi} k^2 + \frac{\kappa_{\text{eff}}}{\chi}.$$

In the long-wavelength limit the phase velocity approaches the shell propagation speed

$$\boxed{c_{\text{shell}}^2 = \frac{\lambda}{\chi}}.$$

This is the first key result: GEM naturally contains a characteristic causal speed with the correct formal role to be compared with the observed speed of light.

3 Depth-Weighted Effective Potentials and Local Stiffness

To let persistence depth modulate local response, consider depth-weighted effective potentials of the form

$$V(\phi, D(x)) = D(x)\, U(\phi - \phi_0),$$

where U is a dimensionless local potential shape. A concrete quartic example is

$$V(\phi, D(x)) = \frac{D(x)}{2}(\phi - \phi_0)^2 + \frac{\alpha}{4} D(x)(\phi - \phi_0)^4 + \cdots .$$

The local effective stiffness at the minimum is then

$$\kappa_{\text{eff}}(x) = \left. \frac{\partial^2 V}{\partial \phi^2} \right|_{\phi=\phi_0} = D(x)$$

in the natural normalisation where $U''(0) = 1$.

Thus, in this class of depth-weighted potentials, deeper regions are locally stiffer in direct proportion to persistence depth. This gives a true spatially varying stiffness:

$$\boxed{\kappa_{\text{eff}}(x) = D(x)}.$$

Depth-Dependent Propagation Structure

Promoting the response coefficients to depth-dependent functions yields a local propagation speed

$$c^2_{\text{shell}}(D) = \frac{\lambda(D)}{\chi(D)}.$$

For weak depth variations $D(x) = D_0 + \delta D(x)$, this induces a spatially varying propagation speed, providing a natural route to refractive or travel-time interpretations of gravitational attraction.

4 Newtonian-Like Attraction from Refractive Propagation

A useful weak-field route treats the shell as an inhomogeneous medium with refractive index

$$n(x) = \frac{c_0}{c_{\text{shell}}(x)}.$$

Rays extremise travel time according to Fermat's principle. If deeper regions correspond to reduced local propagation speed (higher refractive index), trajectories bend toward them, yielding Newtonian-like attraction in the weak-field limit.

5 Effective Metric from Split Response Functions

To capture gravitational redshift and the full weak-field light-bending structure, distinguish temporal and spatial response functions explicitly. Consider the weak-field effective metric ansatz

$$ds^2 = -A(D)\, dt^2 + B(D)\, d\mathbf{x}^2.$$

For weak depth variations this can be expanded to recover the standard isotropic weak-field Schwarzschild form

$$ds^2 \approx -\left(1 + \frac{2\Phi}{c_0^2}\right) dt^2 + \left(1 - \frac{2\Phi}{c_0^2}\right) d\mathbf{x}^2,$$

including the characteristic factor of two in light deflection.

6 Conclusion

This paper has assembled a unified weak-field GEM programme in which propagation speed, local stiffness, and gravitational behaviour all emerge from persistence structure within the coherence shell. The main results are: - A natural shell causal speed $c^2_{\text{shell}} = \lambda/\chi$, - Depth-dependent local stiffness $\kappa_{\text{eff}}(x) = D(x)$, - Newtonian-like attraction from depth-modulated propagation structure, - Weak-field Schwarzschild-like behaviour from split temporal and spatial response functions.

The broader conceptual implication is that internal structure and external dynamics may not be governed by separate principles. They may instead be different expressions of a single persistence architecture. In that sense, gravity is not added to the shell from outside. It is one of the ways the shell behaves when persistence varies across itself.

Future work will focus on rigorous derivations from the microscopic MT/GCF action, numerical simulations of depth-gradient dynamics, and confronting the predictions with observational data.